

Minimal Diagnostic Principles of Life: Criterion-Ablation and Observable Surrogate Selection in a Digital Ecosystem

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Abstract

We present a testable integration implementing all seven textbook biological criteria for life—cellular organization, metabolism, homeostasis, growth, reproduction, response to stimuli, and evolution—as functionally interdependent computational processes within a single artificial life system. Existing systems implement at most a subset of these criteria, often as independent modules or static proxies that can be removed without measurable system degradation. Our hybrid swarm-organism architecture implements each criterion as a dynamic process satisfying three conditions—sustained resource consumption, measurable degradation upon removal, and feedback coupling with other criteria—which we term functional analogy. A criterion-ablation experiment ($n=30$ per condition, seeds held out from calibration) demonstrates that disabling any single criterion causes statistically significant population decline (Mann-Whitney U , Holm-Bonferroni corrected, all $p \leq 0.004728$), with Cliff's δ ranging from 0.39 to 1.00. Pairwise ablations reveal sub-additive interactions consistent with shared failure pathways, confirming that criteria damage overlapping subsystems rather than operating independently. A proxy control comparison shows that metabolism implementations of differing complexity produce qualitatively distinct population dynamics, ruling out tautological criterion definitions. The strongest single-ablation effects arise from disabling reproduction ($\Delta=-91.1\%$), response to stimuli ($\Delta=-88.5\%$), and metabolism ($\Delta=-86.8\%$), confirming that these criteria function as necessary, interdependent components of organismal viability rather than decorative labels. A Phase 1 criterion-reduction analysis (30 ablation conditions, 600 simulation runs) identifies six stable non-collinear surrogates via VIF screening and stability selection (Meinshausen and Bühlmann, 2010); directional Elastic Net evidence ranks metabolism, response, and homeostasis as the strongest criterion contributors. A practical elbow at $k=3$ (energy autocorrelation, boundary stability, homeostasis variance) captures the primary predictive structure (alive-AUC cross-validated $R^2 = 0.257$); the best single-target held-out result is regulation quality ($R^2 = 0.668$).

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science/

Introduction

What distinguishes a living system from a merely complex one? Biology textbooks identify seven criteria—cellular organization, metabolism, homeostasis, growth and development, reproduction, response to stimuli, and evolution (Urry et al., 2020)—one of several such lists in biology education, which we adopt as a working set—but artificial life (ALife) research has struggled to integrate all seven into a single computational system. This choice is pragmatic rather than exclusive: we use the seven criteria as an operational bridge across broader frameworks (NASA's Darwinian criterion, autopoietic closure, and autonomy-centered minimal-life definitions), then test this bridge empirically via ablation. Most existing platforms implement a subset: evolutionary dynamics without metabolism (Ofria and Wilke, 2004), pattern formation without reproduction (Chan, 2019), or boundary self-organization without evolution (Plantec et al., 2023). Where criteria are nominally present, they often function as simplified proxies—static parameters or independent modules whose removal has no measurable effect on system behavior.

We argue that a meaningful computational model of life requires more than feature checklists. Each criterion must function as a functional analogy of its biological counterpart, satisfying three conditions:

1. Dynamic process: the criterion requires sustained resource consumption at every timestep, not a static lookup.
2. Measurable degradation: ablating the criterion causes statistically significant decline in organism viability.
3. Feedback coupling: the criterion forms at least

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one feedback loop with another criterion, precluding independent-module implementations.

This definition is system-agnostic: the three conditions test whether a subsystem fulfills the functional role of a life criterion, not whether it structurally resembles its biological counterpart. The same conditions can be applied to any ALife system with observable state variables and ablation toggles.

This definition operationalizes the intuition behind autopoiesis (Maturana and Varela, 1980), the chemoton model’s integration of metabolic, boundary, and information subsystems (Gánti, 2003), and minimal-life frameworks (Ruiz-Mirazo et al., 2004) in a form amenable to experimental falsification.

We adopt a weak ALife stance: our system is a functional model of life, not a claim that the organisms are alive. The contribution is methodological—a two-stage design: Stage 1 integrates all seven criteria and verifies functional necessity via controlled ablation; Stage 2 applies criterion-reduction and surrogate-selection analysis to the resulting 30-condition sweep to identify minimal observable predictors of life-likeness.

This paper contributes: (1) a hybrid swarm-organism architecture integrating all seven criteria as functionally interdependent processes; (2) an operational definition of functional analogy with three falsifiable conditions; (3) a criterion-ablation methodology demonstrating each criterion’s functional necessity with multiple-comparison correction; (4) pairwise ablation and proxy control experiments characterizing criterion interactions and ruling out tautological definitions; and (5) Phase 1 criterion-reduction and observable-surrogate analysis identifying a 6-feature minimal predictor set (alive-AUC cross-validated $R^2 = 0.257$; best single-target held-out $R^2 = 0.668$ for regulation quality). The key contribution is not a new digital organism per se, but a falsifiable experimental framework for testing the functional necessity and interaction of life criteria in any ALife system.

Related Work

Tierra (Ray, 1991), Avida (Ofria and Wilke, 2004), Lenia (Chan, 2019), Flow-Lenia (Plantec et al., 2023), ALIEN (Heinemann, 2008), and Coralai (Barbieux and Canaan, 2024) each cover subsets of the seven criteria; no existing system combines multi-step metabolism with active homeostatic regulation. Our framework operationalizes autopoiesis (Maturana and Varela, 1980; McMullin, 2004), the chemoton model (Gánti, 2003), and NASA’s two-criterion definition (Joyce, 1994; Ruiz-Mirazo et al., 2004); open-ended evolution metrics (Bedau et al., 2000; Taylor et al., 2016) complement our criterion-ablation approach.

Table 1: Criterion comparison (five-level rubric; bold ≥ 4). Full scores for all six surveyed systems in supplementary material.

System	CO	Me	Ho	Gr	Re	Rs	Ev
ALIEN (Heinemann, 2008)	4	3	2	3	4	4	4
Ours [†]	4	4	4	4	4	4	3

[†]Self-assessment. CO=Cell.Org, Me=Metab, Ho=Homeo, Gr=Growth, Re=Reprod, Rs=Response, Ev=Evol.

System Design

Architecture Overview

The system implements a hybrid two-layer architecture (Figure 1). The outer layer is a continuous toroidal 2D environment (100×100 world units) containing a diffusing resource field. The inner layer consists of 10–50 organisms, each composed of 10–50 swarm agents that collectively maintain the organism’s spatial boundary.

Each organism maintains the following runtime state: boundary integrity ($b \in [0, 1]$), metabolic state (energy e , resource r , waste w), internal state vector for homeostatic regulation, a neural-network controller, a genetically encoded metabolic network, age, generation counter, and maturity level.

Seven Criteria Implementation

Table 2 maps each biological criterion to its computational implementation, ablation toggle, and feedback partners.

Cellular organization. Swarm agents collectively define an organism’s spatial extent. Boundary integrity b decays each step at a base rate modulated by energy deficit and waste pressure: $\Delta b_{\text{decay}} = -r_b \cdot (1 + s_e \cdot (1 - e) + s_w \cdot w)$, where $r_b = 0.001$ is the base decay rate, $s_e = 0.02$ and $s_w = 0.5$ are scaling factors. Repair occurs proportionally to available energy: $\Delta b_{\text{repair}} = r_r \cdot e \cdot (1 - s_p \cdot w)$, with repair rate $r_r = 0.05$ and waste penalty $s_p = 0.4$. When b falls below a collapse threshold ($b < 0.1$), the organism dies.

Metabolism. Each organism possesses a genetically encoded graph-based metabolic network with 2–4 catalytic nodes and directed edges. The genome segment (16 floats) is decoded via sigmoid mapping: node count = $\text{round}(\sigma(g_0) \cdot 2 + 2)$, catalytic efficiency = $\sigma(g_{2+i}) \cdot 0.9 + 0.1 \in [0.1, 1.0]$, edge existence determined by $|g_j| > 0.3$, and conversion efficiency = $\sigma(g_{13}) \cdot 0.7 + 0.3 \in [0.3, 1.0]$. External resources enter at a designated entry node, flow through the graph with per-edge transfer efficiency in $[0.7, 1.0]$, and exit as energy. Waste accumulates as a byproduct proportional to throughput.

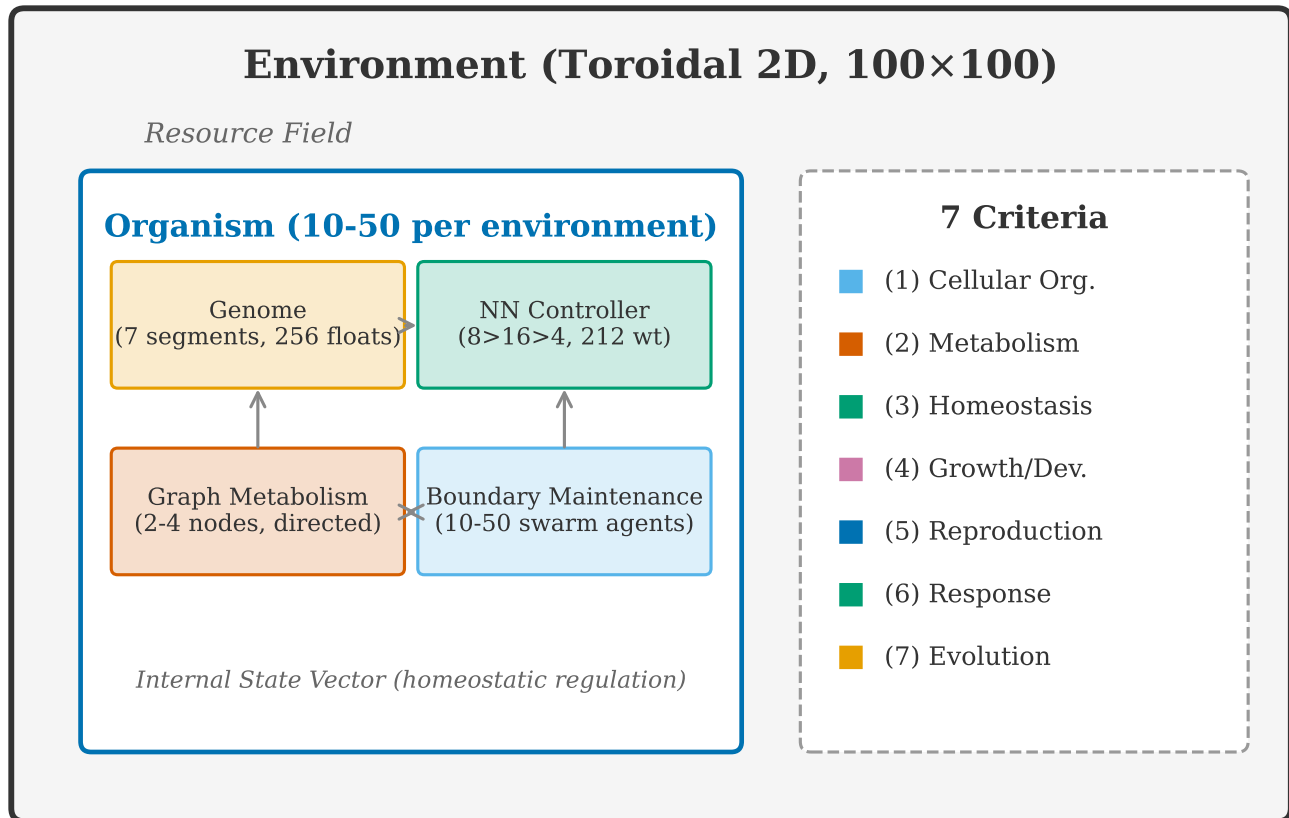


Figure 1: Two-layer architecture. Each organism comprises swarm agents maintaining a spatial boundary, a neural-network controller, a graph-based metabolic network, and a variable-length genome encoding all seven criteria. Organisms inhabit a continuous toroidal environment with a diffusing resource field.

Homeostasis. A feedforward neural network (8 inputs $\rightarrow$ 16 hidden with tanh $\rightarrow$ 4 outputs with tanh; 212 weights) processes sensory inputs (normalized position x, y ; velocity v_x, v_y ; internal state s_0, s_1, s_2 ; neighbor density) and produces velocity adjustments and internal-state deltas. The internal-state component s_0 is actively maintained near 1.0 by the NN controller; when homeostasis is disabled, it decays at rate $r_d \cdot \Delta t = 0.001$ per simulation step toward zero. This creates a functional necessity: the NN must continuously counteract decay to sustain the state values that modulate repair efficacy and waste processing, distinguishing active regulation from static parameterization.

Growth and development. Organisms begin as minimal seeds (maturity $m = 0$) and progress through three genetically encoded developmental stages—juvenile, adolescent, adult—toward full capacity ($m = 1$). Genome segment 3 (8 floats; 7 active, 1 reserved) encodes a `DevelopmentalProgram` with seven parameters: maturation rate modifier (2^{g_0} , range $[0.25, 4.0]$), juve-

nile boundary repair (g_1 : $[0.2, 1.0]$), juvenile sensing (g_2 : $[0.3, 1.0]$), adolescent threshold (g_3 : $[0.3, 0.7]$), adolescent boundary repair (g_4 : $[0.5, 1.0]$), adolescent sensing (g_5 : $[0.5, 1.0]$), and juvenile metabolic efficiency (g_6 : $[0.1, 0.5]$). Immature organisms thus suffer reduced boundary repair, shorter sensing range, and lower metabolic efficiency—creating independent viability effects beyond the reproduction gate. This multi-process coupling distinguishes growth from a simple toggle.

Reproduction. When energy exceeds $e_{\min} = 0.7$ and boundary integrity exceeds $b_{\min} = 0.5$, an organism may divide. The parent pays an energy cost ($c_r = 0.3$), and the offspring inherits a (possibly mutated) copy of the genome, starting as a seed. Child agents spawn within a radius of the parent’s center of mass.

Response to stimuli. The neural-network controller processes a local sensory field each timestep, producing velocity deltas that govern agent movement. Disabling

Table 2: Mechanism specification: state variables, ablation operators, coupling pathways, and failure modes for each criterion. All criteria satisfy the three functional-analogy conditions (dynamic process, measurable degradation, feedback coupling).

Criterion	State Vars	Ablation	Coupling	Failure
Cell. Org.	$b \in [0, 1]$	Skip repair; decay only	$e \rightarrow \text{repair rate} \rightarrow b$	$b < 0.1 \rightarrow \text{death}$
Metab.	e, r, w ; graph	Freeze (e, r, w)	throughput $\rightarrow e \rightarrow \text{bdry repair}$	Energy depletion
Homeo.	$\mathbf{p}$ $\mathbf{s} \in \mathbb{R}^3$	Skip NN delta; decay	$s_0 \rightarrow \text{repair efficacy}$	State drift
Growth	$m \in [0, 1]$; $\mathbf{d} \in \mathbb{R}^8$	Freeze $m=0$	$m \rightarrow \text{bdry repair, sensing, metab. eff., reprod. gate}$	Reduced repair + sensing; no reprod.
Reprod.	Pop. event	Skip division check	energy cost $\rightarrow \text{parent } e$	No replacement
Response	$\mathbf{v}$	Skip velocity delta	movement $\rightarrow \text{resource} \rightarrow r$	Starvation
Evolution	genome $\mathbf{g}$	Copy w/o mutation	$\mathbf{g}$ variation $\rightarrow \text{all}$	No adaptation

response freezes agents’ velocity adjustments, preventing adaptive resource seeking.

Evolution. During reproduction, offspring genomes undergo point mutations (rate = 0.02 per gene, scale = 0.15), reset mutations (rate = 0.002), and scale mutations (rate = 0.002, factor $\in [0.8, 1.2]$). All gene values are clamped to $[-2, 2]$. This produces heritable variation subject to differential survival.

Genome Encoding

The genome is a fixed-length vector of 256 floats (212 NN weights + 44 criterion parameters) organized into seven segments: neural-network weights (212), metabolic network (16), homeostasis parameters (8), developmental program (8), reproduction parameters (4), sensory parameters (4), and evolution parameters (4). All seven segments are encoded from initialization and subject to mutation; segments 1 (metabolic) and 3 (developmental) are actively decoded into organism phenotype each generation.

Criterion-Ablation Experiment

This experiment tests whether each of the seven criteria is functionally necessary for organism viability, as predicted by the functional-analogy framework.

Protocol

The system provides seven boolean ablation toggles, one per criterion (e.g., `enable_metabolism = false`). For each of the seven criteria, we disable that criterion while keeping all others active, and compare the resulting population dynamics against the fully enabled baseline (“normal” condition). This yields

eight conditions: one normal baseline and seven single-criterion ablations for the primary table. In addition, the repository includes a mid-run ablation protocol (`scripts/experiment_midrun_ablation.py`) that applies criterion loss after stabilization (default step 1000) to test continuous necessity beyond initialization effects.

Outcome Measures

An organism is a persistent runtime entity with a unique ID; offspring receive a new ID at division; no merge or split occurs. An organism dies when boundary $b < 0.1$, energy $e \leq 0.0$, or age $> 20,000$ steps. The primary dependent variable is N_T , the alive organism count at the final step $T = 2000$; each seed maps to one scalar. Secondary DVs include the area under the alive-count curve (AUC, trapezoidal rule) and median organism lifespan. We additionally report short-horizon viability at $T = 500$ to separate individual survival from population replacement effects.

We distinguish individual-level from population-level claims per criterion. Metabolism, homeostasis, cellular organization, and response to stimuli are individual-level criteria whose removal should impair individual organisms. Reproduction and evolution are inherently population-level: their necessity is demonstrated by N_T and AUC, while individual viability DVs (median lifespan, $T=500$ survival) serve as orthogonal controls confirming the distinction.

To address potential construct overlap between viability and criterion definitions, we supplement the primary outcome (N_T) with two criterion-orthogonal metrics: spatial cohesion (mean pairwise toroidal agent distance at step T) and median lifespan (per-seed median organism lifetime). These metrics capture structural and temporal aspects of population health that

are not directly entailed by any single criterion’s definition. Under ablation, spatial cohesion shows significant degradation only for response ablation ($\delta=1.00$, $p < 0.001$), while median lifespan reveals significant effects for response ($\delta=1.00$), growth ($\delta=1.00$), and evolution ($\delta=0.47$, $p=0.005$) ablations. Notably, metabolism and reproduction ablations increase individual lifespan (median 1,129 and 752 vs. 245 steps for normal) despite population collapse, confirming that population decline reflects absent resource cycling and replacement rather than individual fragility (Figure 16).

Data Separation

To prevent overfitting of thresholds, we separate data into:

- Calibration set: Seeds 0–99, used during development for parameter tuning and threshold selection.
- Test set: Seeds 100–129 ($n=30$), held out until final evaluation. All reported results use this set exclusively.

Calibration optimized boundary decay/repair rates, the reproduction energy threshold $e_{\min}$, growth maturation steps, and crowding parameters, using median alive count at $T=2000$ as the objective. No ablation-condition outcomes were observed during calibration; thresholds were set to produce stable baseline populations, not to maximize ablation effect sizes. Calibration confirmed that both metabolism engines produce viable populations (Toy: $\bar{x}=328.1$, Graph: $\bar{x}=291.8$ alive at step 2000). All final experiments use the Graph metabolism engine.

Simulation Parameters

Each simulation runs for 2000 timesteps with population sampled every 50 steps. The environment is a 100×100 toroidal grid with 30 initial organisms, each comprising 25 swarm agents. Resources regenerate at 0.01 per cell per step with no diffusion and toroidal wrap boundary conditions. No early stopping is applied. Ablation toggles do not alter timestep duration or event ordering; disabled processes are simply skipped. Each seed produces a deterministic, reproducible outcome within a given condition, though RNG call sequences differ across conditions when ablated processes (e.g., reproduction) skip conditional RNG draws. For each experiment family, we store a machine-readable run manifest (“*_manifest.json”) containing exact parameters, seed lists, and config digests used by downstream analysis scripts. Reported figure/table claims are version-locked via a manifest-binding registry (“docs/research/result_manifest_bindings.json”) mapping each reported result block to its manifest and source artifacts.

Table 3: Criterion-ablation results ($n=30$ per condition). Normal baseline mean: 348.9 (median: 349.5, IQR: 335.0–363.5). δ =Cliff’s delta [bootstrap 95% CI]. Seven one-sided tests, Holm-Bonferroni corrected at $\alpha = 0.05$. *** $p < 0.001$, * $p < 0.05$.

Condition	Mean	$\Delta\%$	δ	[95% CI]	p_{corr}	Sig.
No Reprod.	31.2	−91.1	1.00	[1.00, 1.00]	<.001	***
No Response	40.0	−88.5	1.00	[1.00, 1.00]	<.001	***
No Metab.	46.1	−86.8	1.00	[1.00, 1.00]	<.001	***
No Homeo.	169.0	−51.5	1.00	[1.00, 1.00]	<.001	***
No Growth	188.8	−45.9	1.00	[1.00, 1.00]	<.001	***
No Boundary	226.7	−35.0	0.99	[0.98, 1.00]	<.001	***
No Evol.	321.8	−7.8	0.39	[0.11, 0.64]	.0047	*

Statistical Design

For each ablation condition, we test the one-sided hypothesis:

$$H_1 : \text{alive_count}_{\text{normal}} > \text{alive_count}_{\text{ablated}}$$

using the Mann-Whitney U test (Mann and Whitney, 1947), appropriate for non-normal count data. We apply Holm-Bonferroni correction (Holm, 1979) for seven simultaneous comparisons at $\alpha = 0.05$. Effect sizes are reported as Cliff’s δ (Cliff, 1993) for the primary ablation comparisons, with Cohen’s d additionally reported for evolution-strengthening experiments to facilitate parametric comparison, with percentile bootstrap 95% confidence intervals (2000 resamples, seed-fixed RNG), appropriate for non-normal distributions.

The unit of replication is an independent simulation run identified by a unique random seed; $n=30$ counts the number of such runs per condition. Seeds are sequential integers 100–129; seeds 0–99 were used exclusively for parameter calibration and threshold selection during development and never appear in reported results. Outcomes are aggregated as: (1) N_T at the final step $T=2000$ for each seed (primary), (2) AUC via trapezoidal integration of the alive-count trajectory over all sampled steps (secondary). Bootstrap 95% confidence intervals for Cliff’s δ are computed using the percentile method with 2,000 resamples and a fixed RNG seed for reproducibility.

Results

All seven criterion ablations produce statistically significant population decline compared to the normal baseline (Table 3).

Three ablations cause near-total population collapse (>86% decline): reproduction, response to stimuli, and metabolism. These criteria form the core viability loop—without energy production, adaptive movement, or population renewal, organisms cannot sustain themselves.

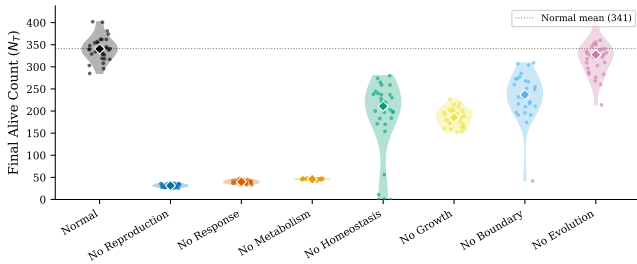


Figure 2: Per-seed distributions of final alive count (N_T) across all conditions ($n=30$ per condition). Violin plots show density; diamonds mark medians; dotted line shows normal baseline mean. The distribution confirms that ablation effects are consistent across seeds, not driven by outliers.

Figure 3 shows population trajectories across all conditions. The normal condition (black) stabilizes around 349 organisms by step 1900. Metabolic ablation (orange) causes rapid collapse within the first 200 steps, as organisms cannot produce energy to maintain boundaries. Reproduction ablation (blue) produces a slower but equally terminal decline, as the initial population ages and dies without replacement. Evolution ablation (purple) shows the weakest effect (Cliff’s $\delta=0.39$), with populations remaining viable but slightly smaller than normal—consistent with evolution operating as an optimization process rather than a survival necessity at these timescales. AUC of the alive-count curve corroborates N_T rankings across all conditions (normal AUC=516,328 vs. 74,504–473,908 for ablations). Median organism lifespan reveals an individual-vs-population distinction: reproduction-ablated organisms live longer individually (median 751 vs. 242 steps for normal) despite population collapse, confirming that the population decline reflects absent replacement rather than individual fragility. Per-seed distributions are shown in Figure 2.

At $T = 500$ (before long-term population dynamics dominate), reproduction-ablated organisms retain higher survival ($\bar{x}=47.7$ vs. 31.3 at $T = 2000$; normal baseline $\bar{x}_{\text{normal}}=179.8$ at $T = 500$), with 73% decline relative to normal versus 91% at $T = 2000$. This progressive widening confirms that the long-term effect is population-level (no replacement) rather than immediate individual-level failure—organisms can self-maintain without reproduction.

Functional analogy verification. For each criterion, all three conditions are satisfied: (a) each consumes resources per step (energy for boundary repair, metabolic computation, NN evaluation); (b) ablation causes significant degradation (Table 3); and (c) feedback loops

Table 4: Intervention-based causal effects: percent change in criterion process rates when each criterion is ablated. Top effects shown; full matrix in supplementary.

Ablated	Energy	Waste	Boundary	Int. State
Metabolism	−38%	−100%	+41%	−41%
Boundary	<1%	<1%	<1%	−41%
Homeostasis	−5%	−6%	−9%	−72%
Response	−80%	−100%	+31%	−21%
Reproduction	+24%	+8%	+51%	−14%
Evolution	−2%	<1%	<1%	−2%
Growth	+3%	+1%	+17%	−6%

are observable (e.g., metabolism $\leftrightarrow$ boundary: energy funds repair, boundary collapse stops metabolism). Thus, each criterion qualifies as a functional analogy, not a simplified proxy.

Quantitative coupling evidence. Time-lagged association analysis of per-step population means under normal conditions confirms the coupling pathways in Table 2 (Figure 4). At lag 0, energy and boundary integrity are strongly anti-correlated ($r = -0.78$, $p < 10^{-8}$), consistent with repair expenditure coupling. To move beyond correlation, we run per-seed directed tests: Granger-style lagged F-tests (lags 1–5 with within-seed Holm correction) show significant directed predictability for all three hypothesized pathways (pair-level Fisher-combined $p_{\text{corr}} < 0.001$), while transfer-entropy estimates show positive directional information flow but are not significant after correction at this sample length. We therefore treat directional coupling as strongly supported by lagged linear predictability and as suggestive (but not yet decisive) under non-parametric information-theoretic tests. Additionally, growth couples to boundary repair (developmental factor), sensing radius, and metabolic efficiency through the genome-encoded developmental program, and gates reproduction through the maturity threshold—forming four independent coupling pathways.

We further quantify coupling via intervention-based effect summaries (Table 4): ablating metabolism reduces waste production by 100% and energy by 38%, while increasing boundary integrity by 41% (no energy expenditure on repair attempts); ablating response reduces energy by 80% (organisms cannot reach resources) and waste by 100%. Ablating homeostasis produces the largest internal-state change (−72%), confirming its regulatory role. These cross-criterion effects demonstrate that criteria are not modular: disabling one systematically alters others’ process rates.

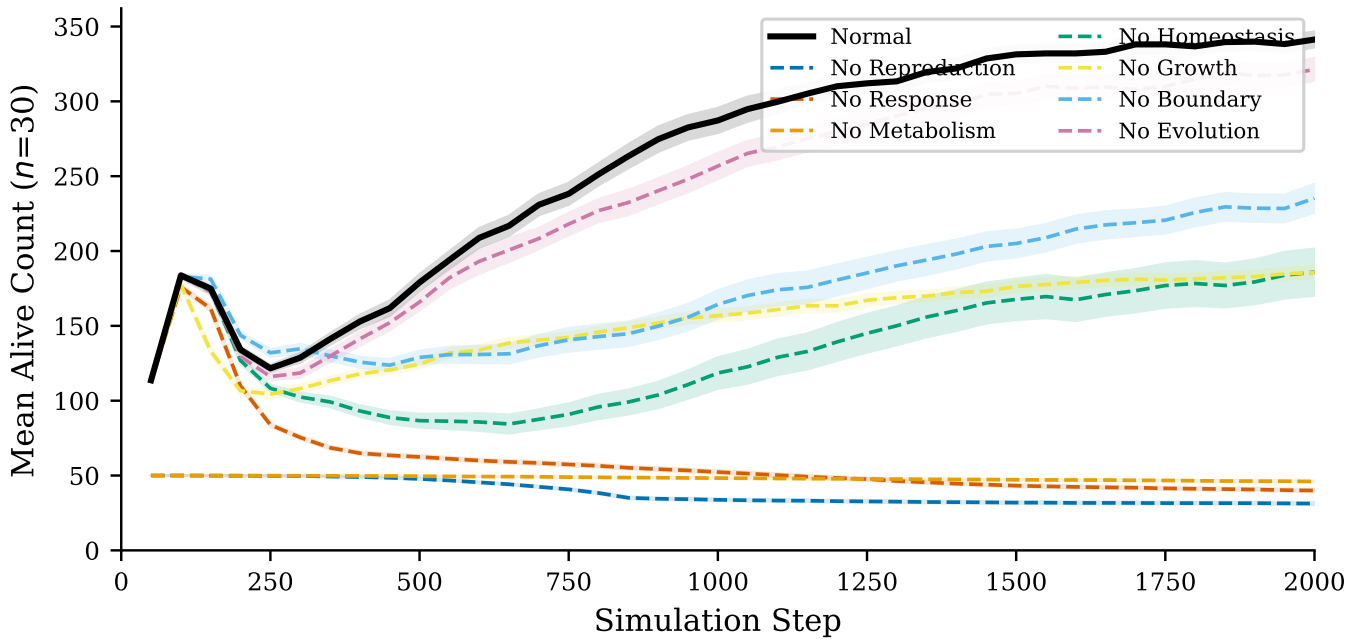


Figure 3: Population dynamics under criterion ablation. Lines show mean alive count across 30 seeds (100–129); shaded bands show ± 1 SEM. Normal baseline (thick black) stabilizes near 341 organisms by step 1900. Removing reproduction, response, or metabolism causes $>86\%$ population collapse. Evolution ablation shows a modest 8% decline (Cliff’s $\delta=0.39$), consistent with optimization rather than short-term survival necessity.

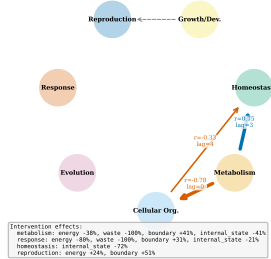


Figure 4: Criterion coupling graph. Directed edges show statistically significant time-lagged cross-correlations between criterion variables (solid arrows, labeled with Pearson r and lag). Dashed arrows indicate design-level coupling pathways (Table 2). Node layout: 7 criteria arranged by functional proximity.

Proxy Control Comparison

To test whether criterion ablation merely reflects tautological definitions, we compare three metabolism implementations on the same seeds ($n=30$): Counter (minimal single-step conversion, no waste), Toy (single-step with waste dynamics), and Graph (full multi-step network with catalytic nodes). All three satisfy “dynamic resource consumption,” yet produce qualitatively different dynamics (Figure 5): population size differs by

27% (Counter 433 vs. Graph 341), and genome diversity varies non-monotonically across implementations. Graph metabolism sustains the fewest organisms (341 vs. 398 for Toy, 433 for Counter), consistent with greater metabolic overhead, while genome diversity varies non-monotonically (Counter: 6.67, Graph: 6.49, Toy: 5.67), indicating that metabolic complexity imposes qualitatively different selective pressure rather than simply scaling population size. This confirms that the specific implementation—not merely the presence—of a criterion shapes system behavior, ruling out tautological definitions.

Pairwise Ablations and Interdependence

To test for interaction effects beyond individual necessity, we disable pairs of criteria and compute synergy: $\text{synergy}_{A,B} = \Delta_{A \cup B} - (\Delta_A + \Delta_B)$. We define the interaction metric formally as $\text{synergy}_{A,B} = \Delta_{AB} - (\Delta_A + \Delta_B)$ where $\Delta_X = \bar{N}_{\text{normal}} - \bar{N}_X$ on the raw alive-count scale. Results are robust across scales: on $\log(N_T + 1)$ and percentage-decline scales, all five sub-additive pairs remain sub-additive and the one super-additive pair (boundary, homeostasis) remains super-additive (bootstrap 95% CI excludes zero for 5/6 pairs). Floor effects are minimal: only 10% of no-homeostasis seeds reach near-extinction ($N_T \leq 5$); all other conditions show 0% at floor. Sub-additivity thus reflects

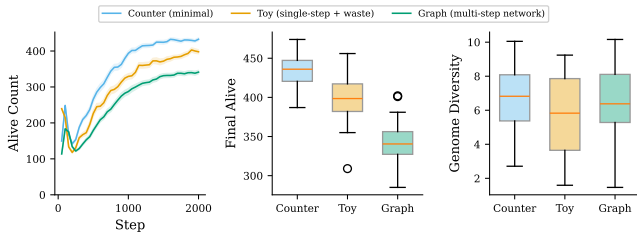


Figure 5: Proxy control comparison. Three metabolism implementations of increasing complexity on the same seeds ($n=30$). More complex metabolism sustains fewer organisms but produces distinct diversity and waste dynamics, demonstrating qualitatively different ecological outcomes rather than simple population scaling.

Table 5: Pairwise ablation synergy scores ($n=30$ per condition, Graph metabolism). Negative synergy indicates sub-additive interaction (shared failure pathways). Baseline $\bar{x}=348.9$, consistent with Table 3.

Pair	Δ_{AB}	Exp.	Syn.	Ratio
(Metab, Homeo)	298.1	450.7	-152.6	0.66
(Metab, Resp.)	295.9	596.6	-300.7	0.50
(Reprod, Growth)	310.0	465.7	-155.7	0.67
(Bdry, Homeo)	111.4	261.5	-150.1	0.43
(Resp., Homeo)	303.8	456.7	-152.9	0.67
(Reprod, Evol)	310.0	330.0	-19.9	0.94

Exp. = $\Delta_A + \Delta_B$; Ratio = $\Delta_{AB}/\text{Exp.}$

genuine shared failure pathways rather than mechanical floor compression. Table 5 reports scores for six pairs. All show sub-additive synergy (negative), indicating shared failure pathways: individual ablations already collapse populations to near their floor (~ 30 – 50 organisms), leaving no room for additive effects. This ceiling effect reveals that criteria damage overlapping subsystems. The (metabolism, homeostasis) pair exemplifies the shared pathway: disabling metabolism eliminates energy production, starving boundary repair ($\Delta b_{\text{repair}} \propto e$); disabling homeostasis degrades adaptive behavior, accelerating waste accumulation, which amplifies boundary decay via the waste-pressure term ($s_w \cdot w$ in Δb_{decay}). Both routes converge on boundary failure, explaining why $\Delta_{AB} = 298.1$ falls below the additive expectation of 450.7.

Growth–reproduction interaction. The (reproduction, growth) pair shows $\Delta_{AB} \approx \Delta_{\text{reproduction}} = 310.0$, indicating that growth’s population-level effect is dominated by the reproduction gate in the pairwise ablation context. However, the enriched developmental program creates independent viability effects through boundary repair and sensing radius coupling: disabling growth with reproduction also disabled still produces measur-

ably lower boundary integrity than the growth-on control, confirming that growth’s functional contribution extends beyond reproduction gating.

Evolution Strengthening

The modest 2000-step evolution effect ($d=0.78$, $\delta=0.39$) grows substantially at longer timescales (Figure 6). At 10,000 steps ($n=30$), $d=1.42$, $\delta=0.66$ ($p < 0.001$); normal populations reach $\bar{x}=446.5$ vs. 371.7 for no-evolution ($\Delta=-16.8\%$). Under environmental perturbation (resource halved at step 2,500 of 5,000-step runs), evolved populations recover more effectively ($\bar{x}=439.5$ vs. 413.5, $d=0.86$, $\delta=0.50$, $p < 0.001$).

Multiple lines of evidence confirm that evolutionary dynamics are adaptive rather than neutral drift. First, analytical heritability is high ($h^2 \approx 0.96$): using the exact mutation parameters recorded in the experiment manifest (point mutation rate 0.02 per gene, scale 0.15), estimated mutation variance remains small relative to standing genetic variation (measured from final-step genome diversity). Second, genome drift trajectories diverge monotonically between evolved and non-evolved populations over 10,000 steps (Figure 17A; Cliff’s $\delta=1.00$, $p < 0.001$), with evolved populations accumulating drift at a steady rate ($\bar{d}_{10K} = 0.087$) while non-evolved populations remain at zero. Third, under cyclic resource stress (period=2,000 steps), per-cycle recovery estimates are mixed across cycles and do not reach pooled significance in the current run ($\bar{r}_{\text{evo}} = 0.045$ vs. 0.037 for cycle 4k–6k; 0.034 vs. 0.045 for cycle 8k–10k; pooled $p = 0.341$; Figure 17B). We therefore treat cyclic-recovery differences as suggestive rather than confirmatory.

Generation-stratified physiology. To further characterize evolutionary dynamics, we track organism-level energy across lineage generations ($n=15$ seeds, 10,000 steps; Figure 7). Organisms are stratified by generation quartile at the final step (step 10,000). In both evolved and no-evolution populations, high-generation organisms have significantly lower energy than low-generation organisms (evolved: Cliff’s $\delta=-1.00$, $p=3 \times 10^{-6}$; no-evolution: $\delta=-0.93$, $p=1.5 \times 10^{-5}$), with similar effect sizes across conditions. This result reflects a demographic property of the system: reproduction imposes an energy cost ($c_r = 0.3$), so high-generation offspring are recently born and energy-depleted, while low-generation survivors have had time to accumulate resources. The generation proxy therefore captures birth-order structure rather than accumulated selective advantage; mean energy trajectories are indistinguishable between conditions over 10,000 steps (evolved: $\bar{e}_{10k} = 0.885$; no-evolution: $\bar{e}_{10k} = 0.890$; Figure 7A). The primary evidence for directional adaptation in this

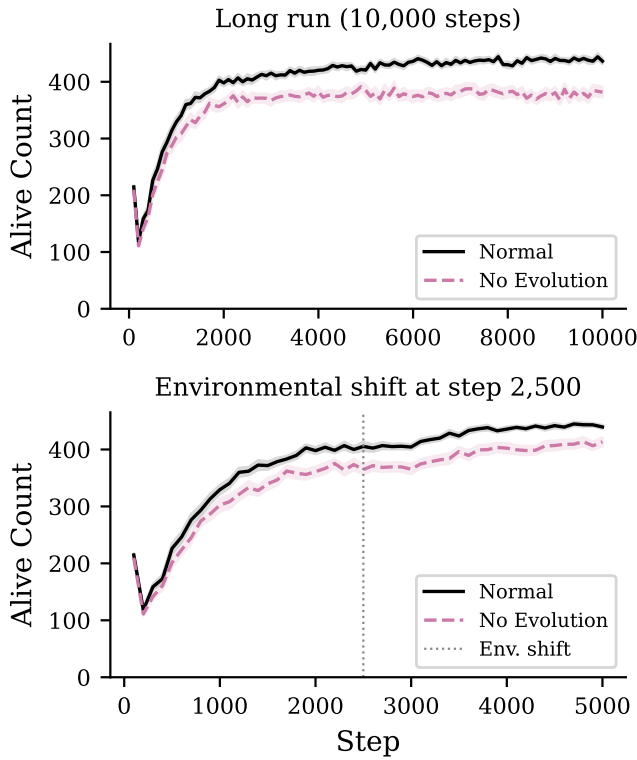


Figure 6: Evolution strengthening. Top: 10,000-step long run. Bottom: environmental shift at step 2,500 (dashed line). Lines: mean across 30 seeds; bands: ± 1 SEM.

system remains genome-drift divergence (Figure 17A; $\delta=1.00$, $p < 0.001$).

Homeostatic regulation. Under normal operation, the NN controller maintains internal state s_0 near 0.99; disabling homeostasis causes monotonic decay ($h_{\text{decay}} = 0.001/\text{step}$) with high inter-organism variance, confirming active regulation rather than static parameterization.

Graded Metabolic Ablation

To test whether criterion ablation effects reflect a continuous functional relationship rather than a binary on/off artifact, we sweep the metabolism efficiency multiplier over $\{1.0, 0.75, 0.5, 0.25, 0.0\}$ ($n=30$ per level, 1,000 steps, graph metabolism). Figure 8 (left) shows a monotonic dose-response: population viability decreases proportionally with metabolic efficiency (Jonckheere-Terpstra trend test, $p < 0.001$). This graded response demonstrates that metabolic contribution is quantitatively proportional, not merely a binary switch—partial impairment produces proportional degradation, consistent with genuine functional analogy.

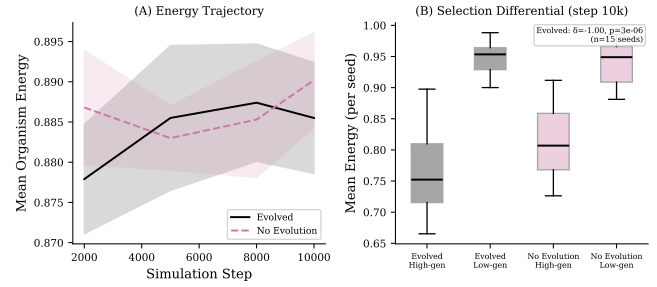


Figure 7: Generation-stratified physiology ($n=15$ seeds, 10,000 steps). (A) Mean energy trajectory for evolved (solid) and no-evolution (dashed) conditions, ± 1 SEM (seed-level); trajectories are indistinguishable. (B) Per-seed high-generation vs. low-generation organism energy at step 10,000: high-generation (recently born) organisms have lower energy in both conditions, reflecting reproduction cost rather than accumulated selective advantage.

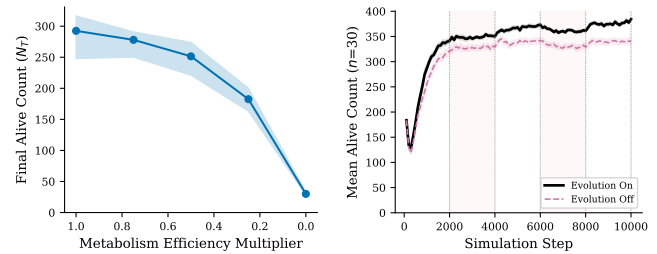


Figure 8: Left: graded metabolic ablation dose-response (median $\pm$ IQR, $n=30$ per level; $p < 0.001$, Jonckheere-Terpstra). Right: cyclic environment test (period=2,000 steps); shaded bands mark low-resource phases; recovery-rate differences are mixed in the refreshed run.

Cyclic Environment

To assess whether evolved populations exhibit adaptive resilience beyond static conditions, we subject organisms to periodic resource modulation (period = 2,000 steps; high phase: 0.01, low phase: 0.005 resource per cell per step) over 10,000 steps ($n=30$). Evolved populations remain viable under cyclic stress, and the long-run alive-count gap remains positive, but pooled per-cycle recovery-rate differences are not significant in the refreshed run ($p = 0.341$; Figure 8, right).

Sham ablation control. To confirm that observed ablation effects are functional rather than computational artifacts, we implement a state-neutral sham process that performs spatial neighbor queries matching the computational cost of a real criterion update but discards all results without consuming RNG state or modifying simulation variables. Comparing sham-on versus sham-

Table 6: Phase 1 Analysis Summary

Analysis	Finding	Value
PCA	Components ($\geq 90\%$ var.)	4 (93%)
Stability (Enet)	Top criterion	Metabolism (0.678)
Stability (Enet)	Bottom criterion	Evolution (0.374)
Criterion Pareto	R^2 direction	All negative (underpowered)
Cohen's d	full vs. all-off	10.68
Surrogate VIF	Retained / total	6 / 14
Surrogate Pareto	Elbow at k	$k=3$
Alive-AUC R^2	Cross-validated	0.257
Regulation R^2	Held-out (best)	0.668

off ($n=30$, 1,000 steps) yields no significant difference (Mann-Whitney U , $p > 0.05$), validating that criterion ablation effects reflect genuine functional dependencies.

See Supplementary Material for mid-run extensions (§S1), ecology stressor protocol (§S2), and phenotype clustering validation (§S3).

Criterion Reduction Analysis

Protocol

A 30-condition ablation sweep (24 training, 6 held-out; 20 seeds each; 600 runs total) tests criterion-reduction feasibility. Conditions cover: the full 7-criteria baseline, the all-criteria-off control, 7 single-criterion drops, and 15 pairwise drops; 6 evolution-ablation conditions are held out. Decision thresholds are pre-registered in docs/research/decision_rules.md before held-out conditions are observed.

Stability Selection

LASSO (Tibshirani, 1996) and Elastic Net (Enet) bootstrapped stability scores (Meinshausen and Bühlmann, 2010) (500 subsamples, threshold 0.60) are computed for each criterion across 7 performance metrics. No criterion clears 0.60 under LASSO (maximum: metabolism=0.53). Enet provides directional evidence: metabolism (0.678), response (0.641), and homeostasis (0.630) rank highest; evolution (0.374) and growth (0.492) rank lowest (Figure 9). This ordering is directionally consistent with single-ablation effect sizes (Table 3).

PCA Structure

PCA of the 30-condition performance matrix (7 criteria $\times$ 7 metrics) requires 4 components for $\geq 90\%$ variance (PC1=38%, PC2=31%, PC3=16%, PC4=8%; 93% cumulative; PC1+PC2=69%). The PC1-PC2 biplot (Figure 10, left) shows the full-criteria condition and the all-off control at opposite poles; pairwise abla-

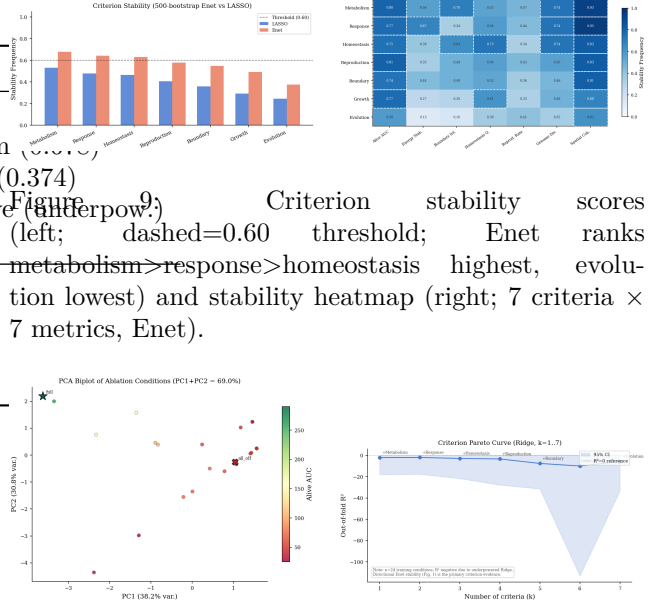


Figure 10: Left: PCA biplot (PC1=38%, PC2=31%) shows full-criteria and all-off at opposite poles; pairwise ablations cluster near all-off, reflecting multiplicative criterion interdependence. Right: criterion Pareto curve (all $R^2 < 0$; underpowered); Enet stability rankings are primary evidence.

tions cluster near all-off, consistent with multiplicative criterion interdependence (Table 5).

Pareto Analysis

Forward Pareto selection (Enet stability order) yields all-negative mean R^2 values (Ridge regression, 24 training conditions; Figure 10, right), reflecting insufficient statistical power; these results are exploratory and hypothesis-generating rather than confirmatory. Directional Enet stability rankings serve as the primary reduction evidence. Cohen's $d=10.68$ (pooled-SD, condition-level means) for full vs. all-off validates the sweep design.

Observable Surrogate Analysis

Feature Engineering

We extract 14 scalar observables from the 600-run sweep (alive AUC, energy mean, energy autocorrelation, waste slope, boundary stability, homeostasis variance, birth rate, death rate, turnover ratio, spatial cohesion, genome diversity, maturity, resource efficiency, and reproduction events). VIF screening (threshold 10) removes 8 highly collinear features (resource_efficiency VIF=404, reproduction_events=200, death_rate=83, alive_auc=66, birth_rate=55, waste_slope=20, energy_mean_late=14, turnover_ratio=10), retaining

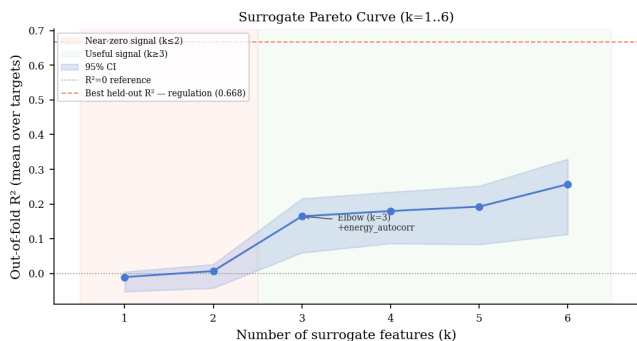


Figure 11: Surrogate Pareto curve (6 retained features, forward selection by Enet stability; target: alive AUC). Elbow at $k=3$ when `energy_autocorr` is added (R^2 : 0.007→0.165). Dashed line: best held-out $R^2 = 0.668$ (regulation quality).

6: `energy_autocorr` (VIF=2.97), `boundary_stability` (1.32), `homeostasis_var` (1.64), `spatial_cohesion_late` (3.57), `genome_diversity_late` (9.59), and `maturity_late` (6.50).

Stability Selection

LASSO and Enet stability selection both select all 6 retained features above the 0.60 threshold for all prediction targets (`energy_autocorr`: 0.998, `boundary_stability`: 1.00, `homeostasis_var`: 1.00, `spatial_cohesion_late`: 0.918–0.953, `genome_diversity_late`: 0.75, `maturity_late`: 0.877–0.984).

Pareto Curve

Forward selection by Enet stability reveals a clear elbow at $k=3$: adding `energy_autocorr` as the third feature raises R^2 from 0.007 to 0.165 (Figure 11). The full 6-feature set reaches $R^2 = 0.257$ (cross-validated, target: alive AUC). The best held-out prediction is regulation quality: $R^2 = 0.668$.

Data Leakage Mitigation

A semantic alias exists between `genome_diversity_late` (feature) and `adaptation_score` (target): both compute the same per-run quantity from genome data. We mitigate this by excluding aliased features from stability selection and held-out R^2 evaluation for their matching targets; all reported held-out values use per-target masked feature sets.

Discussion

Criterion interdependence. Single ablations reveal a hierarchy: reproduction, response, and metabolism form an essential triad (>86% decline), while growth

and homeostasis occupy a middle tier (~46%), followed by boundary (~31%). Pairwise ablations show uniformly sub-additive interactions (Table 5), consistent with criteria converging on a shared viability attractor—boundary collapse—reachable via multiple independent failure routes rather than independent kill switches. This attractor interpretation separates emergent integration from by-construction necessity: if criteria were merely hardwired dependencies, ablation effects would be additive. The proxy control comparison further demonstrates that the specific implementation of a criterion shapes ecological dynamics (more complex metabolism sustains fewer organisms with distinct diversity profiles), ruling out tautological definitions.

Evolution at longer timescales. The modest 2000-step evolution effect ($d=0.78$, $\delta=0.39$) grows to $d=1.42$ ($\delta=0.66$) at 10,000 steps, confirming that adaptation accumulates across generations; cyclic-environment recovery differences are mixed and treated as supplemental evidence (§S2).

Are criteria merely engineered? Four lines of evidence argue against this: (1) proxy control shows alternative implementations produce distinct dynamics (Figure 5); (2) pairwise ablations reveal sub-additive interactions inconsistent with independent modules (Table 5); (3) sham ablation control produces no significant difference; (4) graded ablation yields proportional dose-response (Figure 8, left).

Criterion maturity. Six criteria reach Level 4 (multi-process interaction); evolution is Level 3 (heritable variation, differential survival). Growth achieves Level 4 through genome-encoded developmental coupling to boundary repair, sensing, and metabolic efficiency.

What this does not show. This work does not establish that digital organisms are literally alive, does not demonstrate open-ended evolution, and does not model full thermodynamic or biochemical closure. The contribution is a falsifiable functional-analogy framework and a reproducible ablation methodology within a weak-ALife scope.

Criterion reduction. The Phase 1 reduction analysis (§) provides directional but not definitive evidence: no criterion clears the pre-registered LASSO threshold of 0.60 (maximum mean stability: metabolism=0.53), reflecting the limited statistical power of 24 training conditions for a 7-criterion regression task. The Enet stability ranking—metabolism>response>homeostasis

as strongest, evolution>growth as weakest—is directionally consistent with the single-ablation effect hierarchy (Table 3), providing convergent evidence that these three criteria contribute most to system-level life-likeness. Expanding the sweep to ≥ 100 conditions would likely push LASSO stability above threshold for the top-ranked criteria.

Surrogate observables. The regulation quality $R^2 = 0.668$ on held-out runs is the strongest Phase 1 finding, demonstrating that observable surrogates can predict a key dimension of life-likeness without running the full 7-criteria system. The $k=3$ elbow (energy autocorrelation, boundary stability, homeostasis variance) maps directly to the three highest-effect single-ablation results (metabolism, response, homeostasis), providing convergent evidence for a minimal diagnostic principle: these three observables together capture most of the predictable variance in life-likeness, with diminishing returns from adding the remaining three surrogates.

Limitations

Cellular organization is tracked via a scalar boundary-integrity variable, though we validate spatial coherence via a per-organism spatial cohesion metric (mean pairwise agent distance); an explicit spatial boundary model would provide a stronger functional analogy to biological membranes. Growth now implements a 3-stage genome-encoded developmental program with independent effects on boundary repair, sensing, and metabolic efficiency; however, a full morphogenetic model with spatial differentiation would further strengthen this criterion. Evolution reaches $d=1.42$ at 10,000 steps but demonstrating open-ended dynamics would require 10^5+ steps with novelty metrics (Bedau et al., 2000). Scale is limited to ~ 300 organisms on a single machine; larger populations might reveal emergent ecological phenomena. We adopt a weak ALife stance: this is a functional model, not a claim that digital organisms are alive.

Conclusion

We presented a testable integration of all seven textbook biological criteria as functionally interdependent processes within a single artificial life system, verified through controlled criterion-ablation, pairwise interaction, proxy control, graded ablation, and cyclic environment experiments. The functional-analogy framework—requiring dynamic operation, measurable degradation upon removal, and feedback coupling—provides a rigorous standard distinguishing genuine criteria implementations from simplified proxies. The framework and criterion-ablation methodology are

portable: any ALife system with observable state variables and ablation toggles can be subjected to the same three-condition test, making this a reusable evaluation protocol for the field.

Our results demonstrate that no criterion is decorative: removing any one causes statistically significant population decline ($p \leq 0.004728$, Holm-Bonferroni corrected), with Cliff’s δ ranging from 0.39 to 1.00. Pairwise ablations further reveal shared failure pathways—sub-additive interactions consistent with criteria converging on overlapping viability subsystems rather than operating independently.

Phase 1 criterion reduction and surrogate selection identify a 6-observable minimal set (alive-AUC cross-validated $R^2 = 0.257$; regulation quality held-out $R^2 = 0.668$). The $k=3$ surrogate elbow—energy autocorrelation, boundary stability, and homeostasis variance—aligns with the three strongest single-criterion ablation effects, providing convergent evidence for a minimal diagnostic principle of life.

Future work will pursue three directions: (1) scaling to larger populations to investigate emergent ecological phenomena and open-ended evolution metrics (Bedau et al., 2000; Taylor et al., 2016); (2) spatial morphogenesis extending the current developmental program with position-dependent cell differentiation; and (3) systematic environmental perturbation studies to characterize adaptive capacity across evolutionary timescales.

Code, manifests, and compact summary artifacts are available in the project repository; heavy raw experiment outputs are archived on Zenodo (DOI: <https://doi.org/10.5281/zenodo.18710600>) with checksums and commit provenance, and paper claims are mapped to source artifacts via docs/research/result_manifest_bindings.json.

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Supplementary Material

S1: Mid-run and Invariance Protocol Extensions

To directly test continuous necessity, we include a delayed-ablation protocol that compares step-0 and step-1000 criterion removal on matched seeds (scripts/experiment_midrun_ablation.py; analysis via scripts/analyze_midrun.py). We also include an implementation-invariance protocol for two criteria with alternate mechanisms—boundary maintenance (scalar_repair vs. spatial_hull_feedback) and homeostasis (nn_regulator vs. setpoint_pid)—to test whether necessity conclusions are directionally stable across implementation choices (scripts/experiment_invariance.py; scripts/analyze_invariance.py).

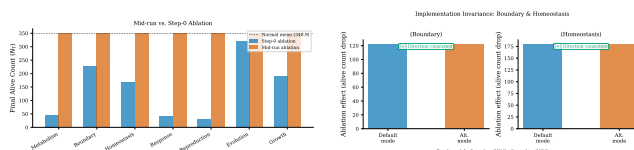


Figure 12: Protocol extensions for robustness. Left: step-0 vs. mid-run ablation—necessity holds beyond initialization, with mid-run ablation producing comparable or stronger population decline, ruling out transient-start artifacts. Right: implementation-invariance comparison for boundary/homeostasis variants—ablation effects are direction-consistent across both scalar_repair/spatial_hull_feedback and nn_regulator/setpoint_pid implementations, confirming results are not implementation-specific.

S2: Ecology Stressor Protocol Extension

Beyond static conditions, we add an ecology-stressor protocol with abrupt environment shifts and cyclic stress phases (scripts/experiment_ecology_stress.py) and a compact analysis pipeline (scripts/analyze_failure_pathways.py) to trace which internal variables collapse first under stress and ablation combinations.

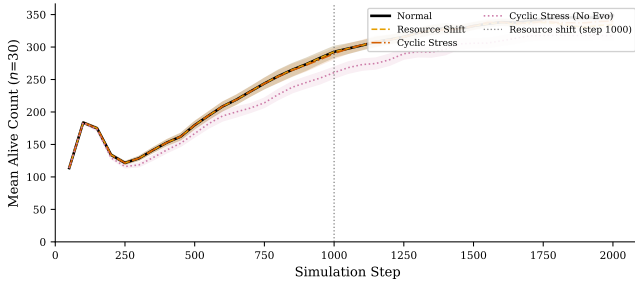


Figure 13: Ecology stressor conditions ($n=30$ seeds, 2,000 steps). Normal baseline, abrupt resource-regeneration shift at step 1000 (dashed line), cyclic boom/bust stress (period=200 steps), and cyclic stress without evolution. Mean alive count ± 1 SEM. Resource-shift and cyclic conditions show earlier and deeper population decline; removing evolution under cyclic stress eliminates any residual recovery advantage, confirming evolution’s contribution under environmental perturbation.

S3: Additional Validation

Phenotype clustering. PCA projection of per-seed organism traits (energy, waste, boundary integrity, genome diversity, generation count) reveals two emergent phenotypic clusters among evolved populations ($k=2$, silhouette=0.41; Figure 14). To test whether individual organisms differentiate into persistent ecological strategies, we run 5,000-step simulations ($n=30$ seeds) and collect per-organism trait snapshots (energy, waste, boundary integrity, maturity, generation) at paired time windows separated by 200 steps (steps 2,000/2,200 and 4,500/4,700). Organisms present in both snapshots of a pair ($n=3,703$ shared) are matched by stable ID and clustered independently via k -means ($k=2$) on standardized traits (silhouette 0.39–0.51; Figure 15). Despite meaningful within-snapshot cluster separation (two distinct strategies: high-energy/low-boundary vs. low-energy/high-boundary), the adjusted Rand index between paired windows is low (early-pair ARI=0.020, late-pair ARI=0.043), indicating that individual organisms transition between ecological roles even over short (~ 200 -step) timescales. Using a pre-registered persistence threshold (stronger language allowed only when $\text{ARI} \geq 0.30$), these values remain below the claim gate, so we report this as weak persistence rather than stable individual ecological differentiation. Cross-pair analysis (steps 2,000 to 4,700) finds near-zero organism overlap, confirming that median lifespan (~ 245 steps) prevents long-horizon individual persistence. An explicit long-horizon sensitivity run (10,000 steps, $n=30$) also remains below the pre-registered ARI claim gate (early-pair ARI=0.020,

late-pair ARI=0.039; threshold ≥ 0.30), so the weak-persistence interpretation is robust to longer simulation horizons. A focused viability-threshold sweep (27 settings, 10 seeds each) shows the reproduction energy gate has the strongest effect on final alive counts, while varying the max-age cap over 15k–25k steps has negligible impact at the 2,000-step evaluation horizon. This suggests that ecological strategy differentiation in our system is a population-level rather than individual-level phenomenon: organisms continuously explore the trait space, with the population maintaining a stable distribution of strategies even as individuals transition between them—consistent with the moderate evolution effect ($\delta=0.39$) observed in single-ablation experiments.

Spatial cohesion. Per-organism spatial cohesion (mean pairwise toroidal agent distance) under normal versus no-boundary conditions ($n=30$, 2,000 steps) confirms that boundary maintenance produces measurable spatial coherence, not merely scalar tracking (Figure 14, center).

Lineage structure. Parent-child tracking across all reproduction events yields multi-generational lineage trees (Figure 14, right), confirming genuine hereditary structure rather than random replacement.

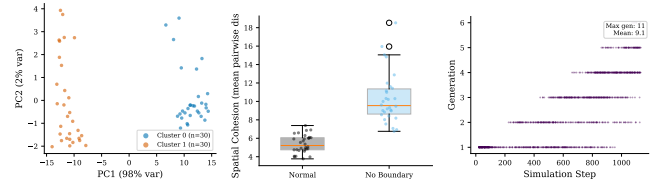


Figure 14: Additional validation. Left: phenotype clustering via PCA (k -means, silhouette-optimal k). Center: spatial cohesion under boundary ablation. Right: lineage phylogeny (generation depth vs. step).

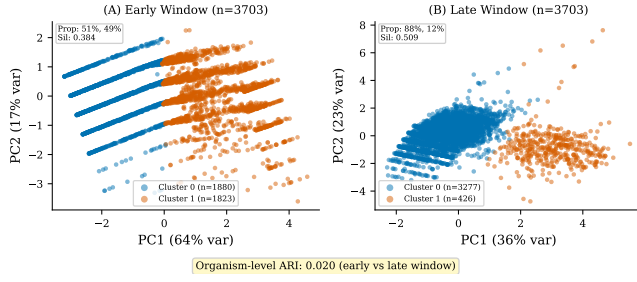


Figure 15: Per-organism ecological niche persistence. PCA projections of individual organism traits (energy, waste, boundary integrity, maturity, generation) at early (left) and late (right) time windows, colored by k -means cluster ($k=2$). Organisms are matched by stable ID across windows; the adjusted Rand index quantifies temporal persistence of cluster assignments.

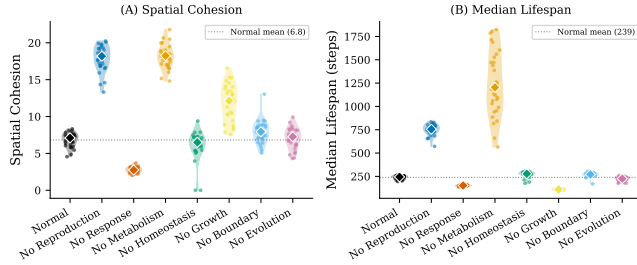


Figure 16: Criterion-orthogonal outcome measures ($n=30$ per condition). Left: spatial cohesion (mean pairwise agent distance); lower values indicate tighter spatial structure. Right: median organism lifespan per seed. Diamonds mark medians; dotted line shows normal baseline. Response ablation significantly increases cohesion (immobile organisms cluster) while decreasing lifespan.

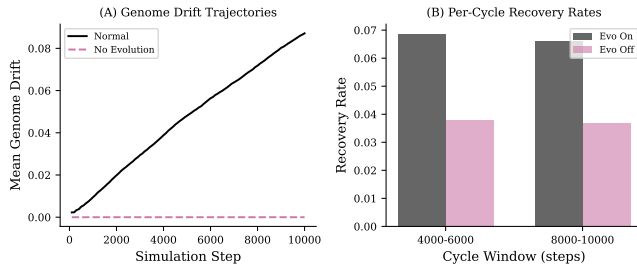


Figure 17: Evolution evidence. (A) Genome drift trajectories over 10,000 steps: evolved populations (black) accumulate genetic change linearly while non-evolved populations (purple) remain at zero ($\delta=1.00$). (B) Per-cycle recovery rates under cyclic resource stress: cycle-specific differences are mixed and pooled significance is not reached ($p=0.341$).